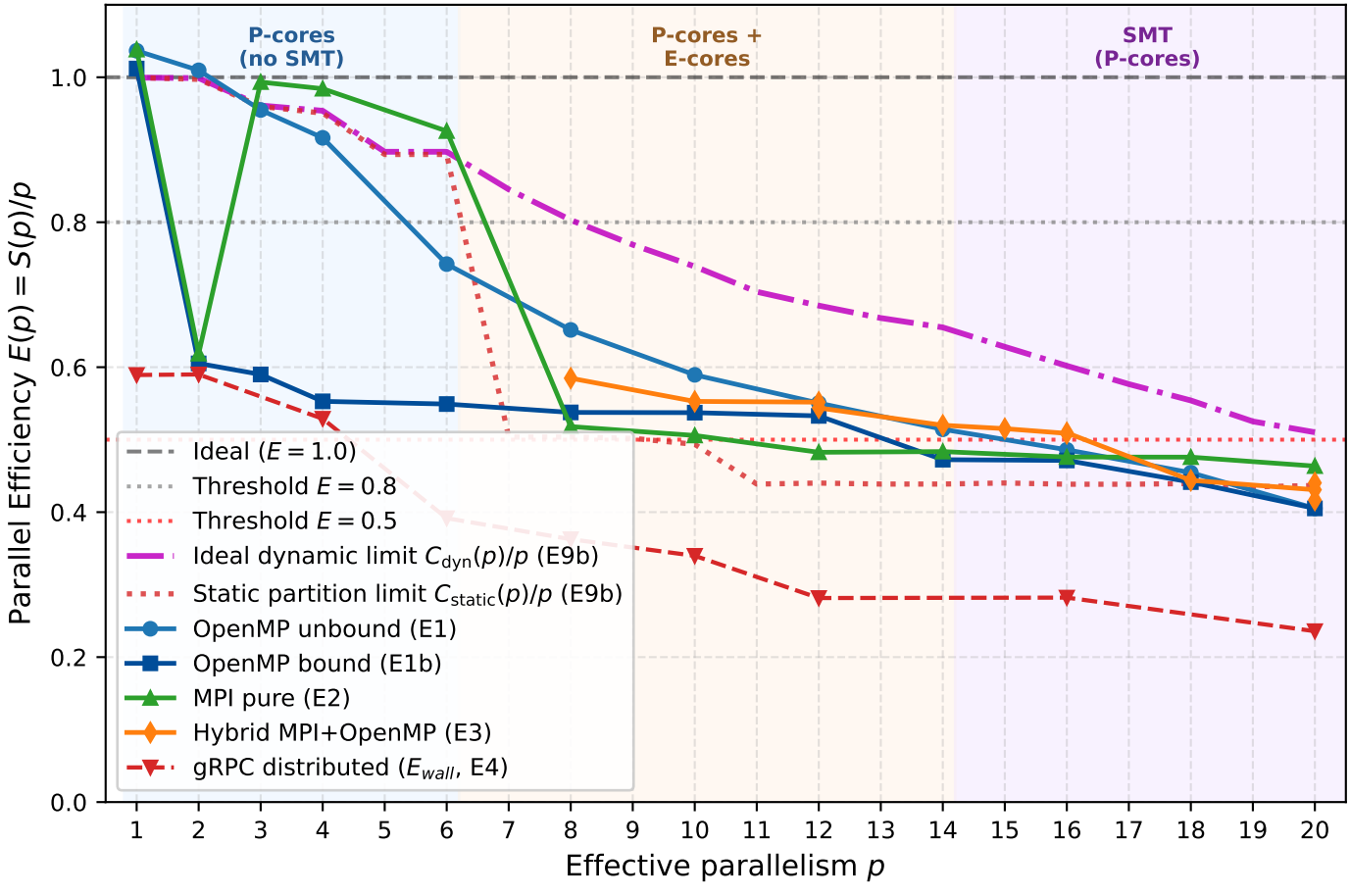
Fig. B — Parallel Efficiency $E(p)$ vs Number of Workers
P-cores (no SMT)
P-cores + E-cores
SMT (P-cores)
Parallel Efficiency $E(p) = S(p)/p$
Effective parallelism p
1.0
0.8
0.6
0.4
0.2
0.0
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Ideal ($E = 1.0$)
Threshold $E = 0.8$
Threshold $E = 0.5$
Ideal dynamic limit $C_{dyn}(p)/p$ (E9b)
Static partition limit $C_{static}(p)/p$ (E9b)
OpenMP unbound (E1)
OpenMP bound (E1b)
MPI pure (E2)
Hybrid MPI+OpenMP (E3)
gRPC distributed (E_{wall}, E4)